CEA Report

ECAT Exam System

Submitted by:

Noor Fatima

Roll No: 63

Section: B

Submitted To:

Hafiz Mubashar Imran

BS Cyber Security

Department of Computer Engineering

UET Lahore

Introduction:

This project is a simple ECAT Exam System developed in Python. The purpose of this project is to provide an exam environment for students and allow administrators to manage the question bank. The system contains two portals:

- Admin portal
- Student portal

The admin portal is used to manage questions and view student results, while the student portal allows students to log in, attempt the exam, and view their results after submission.

Objectives:

The main objectives of this project are:

- To create a simple ECAT examination system.
- To allow students to attempt MCQ-based exams.
- To calculate scores automatically.
- To store student results in memory.
- To allow the admin to manage questions.

Features of the System:

1. **Admin Portal:**

The Admin Portal provides the following features:

- Admin Login
- View Questions
- Add New Questions
- Delete Questions
- View Student Results
- Question bank statics

2. **Student portal:**

The Student Portal provides the following features:

- Student Login
- Start Exam
- Answer Questions
- Skip Questions
- View Final Result
- View Percentage and Grade

Data Structures Used:

The following data structures are used in this project:

1.Lists:

- Questions list stores all exam questions.
- All results list stores student results.

2.Dictionaries:

Each question is stored as a dictionary containing:

- Subject
- Question
- Choices
- Correct Answer

Student results are also stored in dictionaries.

Marking Scheme:

The marking scheme used in the project is:

- Correct Answer = +4 Marks
- Wrong Answer = -1 Mark
- Skipped Question = 0 Marks

The total score is calculated automatically after the exam.

Grade Criteria:

Grades are assigned according to the percentage:

- Excellent = 80% and above
- Good = 65% to 79%
- Average = 50% to 64%
- Below Average = Below 50%

Python Concepts Used:

The following Python concepts were used during the development of this project:

- Variables
- Constants
- Input and Output Functions
- If-else Statements
- Loops
- Functions
- Lists
- Dictionaries
- String Methods
- . append ()
- . pop ()
- len()
- range ()
- import time

Working of the System:

When the program starts, the main menu is displayed. The user can choose either the admin portal or the student Portal.

If the admin portal is selected, the admin enters the username and password. After successful login, the admin can manage questions and view results.

If the student portal is selected, the student enters login information and starts the exam. The student can answer questions or skip them. After completing the exam, the system calculates the score, percentage, and grade automatically and displays the result.

Architecture Diagram:

```
                              ECAT EXAM SYSTEM
                                     |
          -------------------------------------------------------------
            |                                                   |
        Admin Portal                                      Student Portal
            |                                                   |
          Login                                               Login
            |                                                   |
  ----------------------------------------       ----------------------------------
     |          |                |                    |               |
   View    Add Questions    Delete Questions      Start Exam     View Result
 Questions
```

Conclusion:

This project helped me understand the practical use of Python concepts such as functions, lists, dictionaries, loops, and conditional statements. It also improved my problem-solving skills and understanding of menu-driven programs. The project successfully performs the basic functions of an ECAT examination system.